

# VayuNetra

The operations layer for urban air quality.

We don't just measure the air. We **trace** it, **predict** it, and **act** on it — live in ten Indian cities, on ₹0 infrastructure.

## DECEMBER 2025 · WORST DAY OF THE MONTH

**452**

**Delhi**

Severe

**345**

**Chennai**

Very Poor

**331**

**Kolkata**

Very Poor

**302**

**Pune**

Very Poor

Indian National AQI from each city's own CPCB stations. 10 of 10 cities have a December 2025 record.

Team DaGoats · Omkar Kadam · Sejal Kumbhar · Abhinav Prasad

Live: [vayunetra-aqi.vercel.app](https://vayunetra-aqi.vercel.app) · dots = worst day in December 2025 · built 19 Aug 2026



## THE PROBLEM

# The data exists. The layer that acts on it doesn't.

1.67M

premature deaths every year in India  
from air pollution

Lancet Planetary Health (GBD), as cited in the brief

900+

CAAQMS monitoring stations already  
deployed — the data exists

CPCB continuous monitoring network

31%

of monitored cities have **any** response  
protocol

CAG performance audit, 2024

**A reading turns red. Then nothing happens.** No system tells a city officer **who is polluting this square kilometre right now, what the air will be tomorrow, and where to send inspectors first.**

Status quo

reactive orders — 13 of 17 GRAP orders last winter came *after* the AQI had crossed the threshold; weeks to enforcement

VayuNetra

cited recommendation in **seconds**; approval → dispatch → closure timestamped on an audit trail

# One loop: trace → predict → act → protect

## TRACE

Who is to blame, per ~1 km<sup>2</sup> cell — GBM + SHAP blended with chemical-signature priors; abstains where it lacks skill.



Delhi, latest run: traffic 41.9% · transported 21.2% · construction dust 17.2%

## PREDICT

PM2.5 at +24/48/72 h per cell, 80 % conformal band, calibrated P(>120), P(>250).

share of Delhi expected in Very Poor air (calibrated P)



from calibrated P(>120) per cell × population · GET /exposure

## ACT

Ranked worklist → Sentinel-2 evidence + retrieved regulation → draft notice → approve · dispatch · **close with finding**, on an audit trail.

# 413

ranked, cited recommendations across ten cities · draft notice PDF each

live worklist, this morning

## PROTECT

Templated advisories (no LLM) in eight scripts — app, Telegram, IVR line, public displays.

Delhi, Ward 12: air is forecast very poor in +24h. Keep outdoor activity short, use an N95 mask, and move heavy work outside the peak window.

English · Very Poor, +24 h · templated

**Five AI agents on one graph** — signal → cited recommendation in **1.2 s** of compute, measured live; approval, dispatch and closure timestamped per action. The spatial unit is the same ~1 km cell everywhere, so every claim traces from a raw reading to an action.

# Delhi's winter, replayed — against the government's real orders

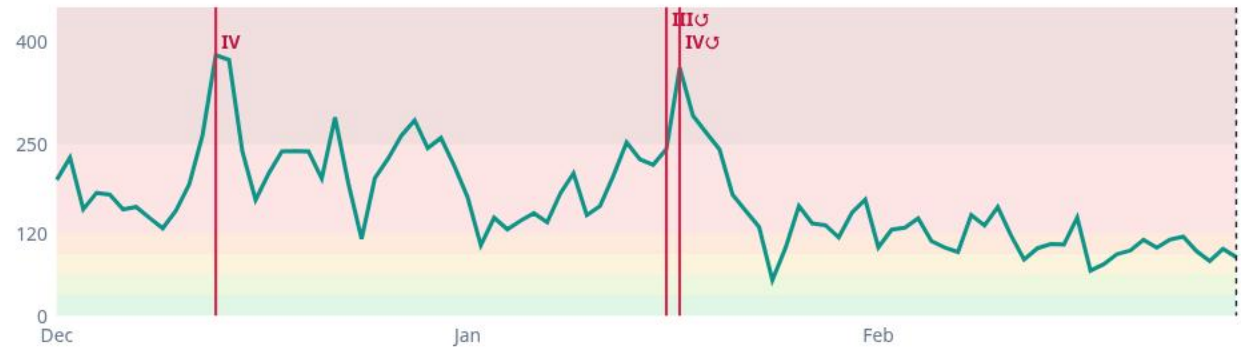
**216.3** Dec 2025 mean  $\mu\text{g}/\text{m}^3$  · AQI 374 Very Poor    **175.3** Jan 2026 mean  $\mu\text{g}/\text{m}^3$  · AQI 343 Very Poor    **106.6** Feb 2026 mean  $\mu\text{g}/\text{m}^3$  · AQI 254 Poor



39 station cells (drawn 2.2x for legibility) · daily mean PM2.5 · CPCB via OpenAQ  
outline = Delhi municipal wards; some CPCB stations sit outside it, in the wider NCR

city-mean PM2.5 ( $\mu\text{g}/\text{m}^3$ ) · daily

28 Feb 2026



**Stage IV** 2025-12-13  
24 h before: **P(>120) 83%** · 99% of 38 cells  
persistence said 225 · at order 382

**Stage III** 2026-01-16  
24 h before: **P(>120) 85%** · 100% of 1 cells  $\triangle$   
persistence said 205 · at order 214

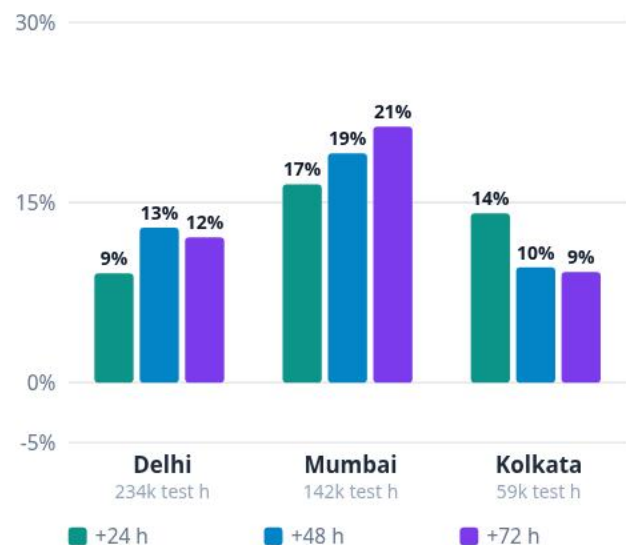
**Stage IV** 2026-01-17  
24 h before: **P(>120) 72%** · 100% of 1 cells  $\triangle$   
persistence said 131 · at order 333

Served forecast replayed with the production recipe (rolling monthly refit, persistence blend, calibrated P). Green =  $P(>120) \geq 30\%$  a day before the order.  $\triangle$  = the public feed carried a single Delhi station that week. Sources: CAQM / PIB releases, linked in docs/OUTCOMES.md.

# Every number is measured — including our failures

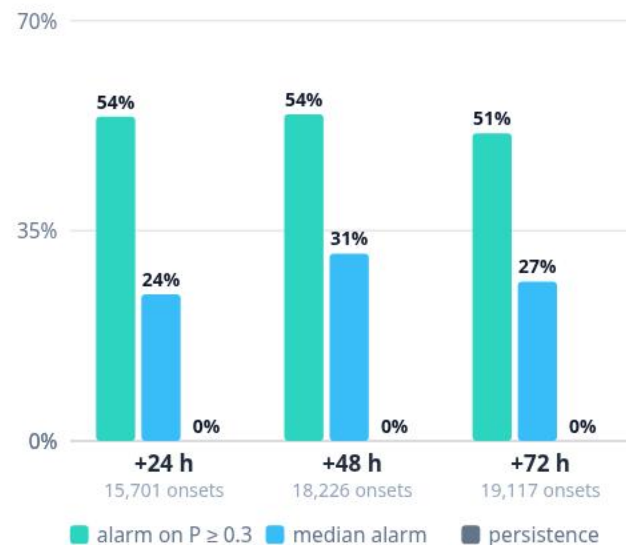
## Served forecast skill vs persistence

$1 - \text{RMSE} / \text{RMSE}_{\text{persistence}}$  · strict temporal split, monthly refit on trailing 90 d



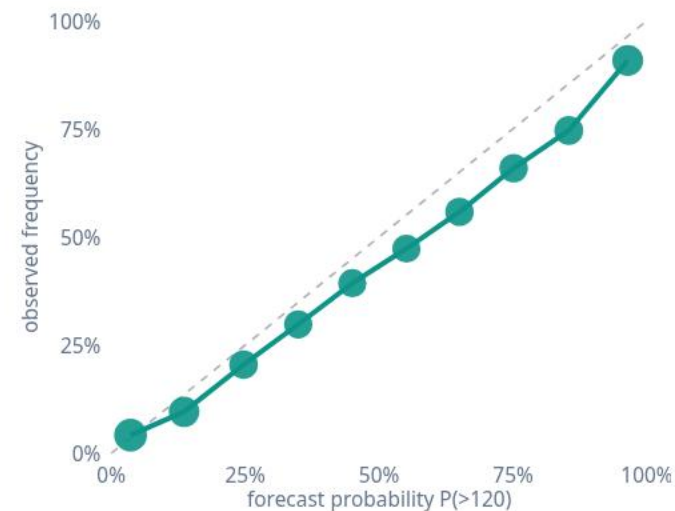
## Warns before the air turns

clean → Very-Poor onsets flagged, alarm on calibrated  $P \geq 0.3$  · Delhi, multi-season



## The probabilities are honest

reliability of  $P(>120)$  at +24 h · 80 % band coverage



• Attribution vs published apportionment: cosine 0.88 / 0.90 / 0.93 · bucket tables vs TERI-ARAI 2018 & Guttikunda 2019 — disagreements stated

• TFT trained on GPU and rejected — LightGBM won held-out skill

• 204 backend tests · 16 e2e flows · CI on every push

LIVE DEMO

# Delhi, right now.

**1 Morning brief → blame map**  
what changed overnight, where the air turns; click the worst hexagon → sources, SHAP, R<sup>2</sup> gate, 72-h forecast with P(Very Poor)

**2 Worklist → evidence → notice**  
real Sentinel-2 image + retrieved regulation → draft notice PDF; Approve · Dispatch · **Close case** · History

**3 Advisories in eight scripts**  
Hindi → Tamil; live Telegram bot; IVR line

**4 Proof, then ten cities**  
validation card · real orders in hindsight · scoreboard → Mumbai

**press D** opens the live console inside this deck · **D** returns · offline: the last screenshot is shown, labelled

# Built to deploy, not to demo

## What it costs to run — measured, not assumed

monthly infrastructure vs cities · one row per reading · 180-day raw retention with archive



Supabase free 500 MB → Pro US\$25; Render free → Starter US\$7 for an always-on API. docs/SCALE.md

## RUNNING AT PRODUCTION SCALE — LIVE

**10**

cities live · Delhi to Lucknow

**16,529**

~1 km<sup>2</sup> cells modelled

**647**

registered + satellite-detected sources

**5,495**

vulnerability-scored zones

**413**

RAG-cited enforcement recommendations

**8**

advisory scripts · app / Telegram / IVR / displays

## Adoption path

- **One YAML per city** (bbox, languages, authority) + one backfill run — 7 metros in a week
- **Closed loop:** approve → dispatch → close with finding, immutable audit trail
- **PRANA-ready export** of every tracked action with its measured effect — feeds the official NCAP portal
- Delhi ROI card: **₹0.91 L cr/yr of Delhi's annual health burden avertable at the ~30 % NCAP target (18,219 deaths/yr) — every figure cited**

India measures. India forecasts.

# VayuNetra operates.

Traced, predicted, cited, delivered, **closed**. In seconds. In eight scripts. For ₹0.



**vayunetra-aqi.vercel.app**

[github.com/omkarr88/VayuNetra](https://github.com/omkarr88/VayuNetra) · @aqivayu\_bot

Omkar Kadam · Sejal Kumbhar · Abhinav Prasad — thank you.



## APPENDIX

# For the Q&A

**A1** Architecture

**A2** Five agents live

**A3** Evidence dossier

**A4** Citizen reach — eight scripts

**A5** Models & validation, all ten cities

**A6** Honesty & fairness

**A7** Cost, security & roadmap

**A8** Ten cities

**A9** Past air

**A10** Positioning vs SAFAR / DSS / PRANA

**A11** Early warning & calibration

**A12** Attribution validation & exposure

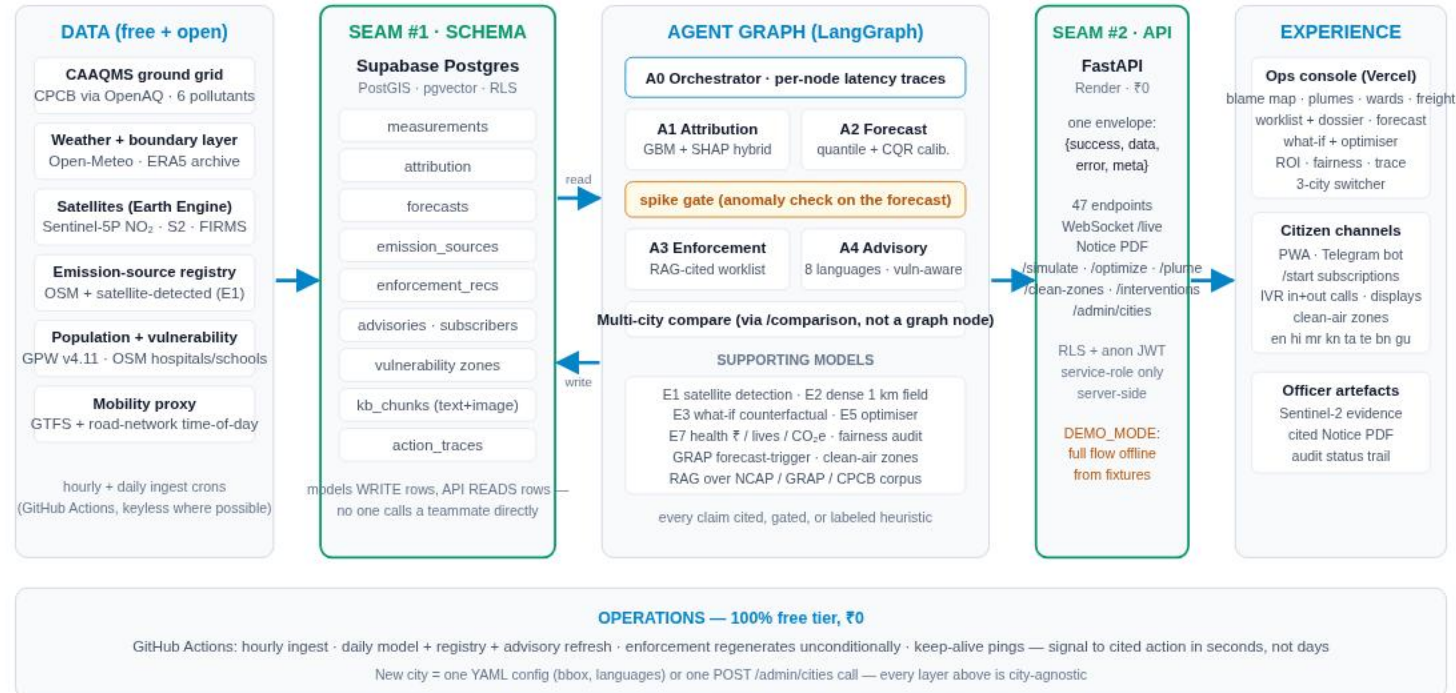
**A13** Real orders in hindsight — did the air change?

**A14** The officer loop closes — audit trail

# Signal to cited action — two seams, one envelope

## VayuNetra — signal to city action in minutes

Every box decouples through two seams: the Supabase schema and the API contract. Spatial unit everywhere: Uber H3 res-8 (~1 km hexagons).



FastAPI (44 routes + WebSocket) on Render · React + MapLibre / deck.gl on Vercel · Supabase Postgres + PostGIS + pgvector with RLS (writes only through the service role) · LangGraph agents · GitHub Actions crons — all free tier for the ten cities running today.

# Five agents on one graph, traced per node

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OPERATIONS

Enforcement

Forecast

Advisories

Cities

Simulator

Impact

Pipeline

Delhi

IN - CPCH

US - EPA

WHO

Present

Tour

Public site

TRUST · DELHI

Pipeline

The six AI agents, run live in front of you — every step traced, every number sourced.

1 Run the agents

2 Read the trace

1 Agent Pipeline

1.2s end-to-end

last multi-agent run: detect → decide → issue

No trace yet — press "Run agents live" to watch the pipeline think.

Run agents live

## Latest trace, Delhi

|                                                             |        |
|-------------------------------------------------------------|--------|
| attribution agent                                           | 242 ms |
| forecast agent                                              | 503 ms |
| enforcement agent — skipped this run (spike gate: no spike) | –      |
| advisory agent                                              | 408 ms |
| signal → cited recommendation                               | 1.2 s  |

action\_traces · signal → cited recommendation is compute time; approval / dispatch / closure are separate timestamped steps

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A2

# A recommendation you can defend

**VayuNetra OPERATIONS**

Enforcement Forecast Advisories Cities Simulator Impact Pipeline

Dethi IN - CPCB US - EPA WHO Present Tour Public site

**Sentinel-2 patch - Satellite-detected construction site #20**  
construction - 86% detection confidence

REGULATORY CITATIONS (RAG)

**Graded Response Action Plan (GRAP)**  
hospitals: additional respiratory care wards activated - Helpline numbers published; mobile health units deployed. match 24%

**Construction dust - Punjabi Bagh**  
~4 km **2k/hr** - 2h - priority 10 - rubric 8/10  
Construction dust contributes approximately 59.7% of PM2.5 in this cell (~15,000 residents exposed). Satellite-detected construction site #75 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).  
60% contribution - 15,000 exposed - [+1 similar site here](#)  
[Evidence dossier](#) [Notice PDF](#) [Approve](#) [Dismiss](#)

**Industrial emissions - Punjabi Bagh**  
~4 km **112/hr** - 8h - priority 9 - rubric 7/10  
Industrial emissions contributes approximately 22.5% of PM2.5 in this cell (~11,854 residents exposed). Rithala Power Plant is the registered industry source at this location. Industrial inspection required: verify stack emission norms, Consent-to-Operate (CTO) compliance, and OCEMS data. Regulatory basis: Graded Response Action Plan (GRAP).  
23% contribution - 11,854 exposed  
[Evidence dossier](#) [Notice PDF](#) [Approve](#) [Dismiss](#)

**Construction dust - Ndmc Charge 3**  
~3 km **24k/hr** - 2h - priority 7 - rubric 8/10  
Construction dust contributes approximately 49.3% of PM2.5 in this cell (~15,000 residents exposed). Satellite-detected construction site #14 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).  
59% contribution - 15,000 exposed - [+12 similar sites here](#)  
[Evidence dossier](#) [Notice PDF](#) [Closed - not applicable - 2026-08-18](#) [History](#)

**Construction dust - Ndmc Charge 3**  
~3 km **6k/hr** - 2h - priority 3 - rubric 8/10  
Construction dust contributes approximately 32.9% of PM2.5 in this cell (~15,000 residents exposed). Satellite-detected construction site #13 is the registered construction source at this location. Site inspection required: verify dust suppression norms compliance (anti-smog gun, water sprinkling, green net coverage). Regulatory basis: Graded Response Action Plan (GRAP).  
33% contribution - 15,000 exposed - [+3 similar sites here](#)  
[Evidence dossier](#) [Notice PDF](#) [Approve](#) [Dismiss](#)

**Real Sentinel-2 imagery** of the actual site (multimodal RAG over image patches + regulatory text).

**Regulation retrieved verbatim** — cited chunk by chunk; the right instrument for the city (GRAP only in Delhi-NCR).

**Draft notice PDF:** addressee, provisions, 24-h deadline, satellite image, projected impact of compliance (screening estimate), provenance block. Stamped **pending officer authorisation**.

## The same advisory, in the reader's own script

### ENGLISH

Delhi, Ward 12: air is forecast very poor in +24h. Keep outdoor activity short, use an N95 mask, and move heavy work outside the peak window.

### HINDI

Delhi, Ward 12: अगले 24 घंटों में हवा बहुत खराब रहने का अनुमान है. बाहर की गतिविधि कम रखें, N95 मास्क पहनें, और भारी काम पीक समय के बाद करें.

### KANNADA

Delhi, Ward 12: ಮುಂದಿನ 24 ಗಂಟೆಗಳಲ್ಲಿ ಗಾಳಿಯ ಗುಣಮಟ್ಟ ತುಂಬಾ ಕಳಪೆ ಇರಲಿದೆ ಎಂದು ಅಂದಾಜಿಸಲಾಗಿದೆ. ಹೊರಗಿನ ಚಟುವಟಿಕೆ ಕಡಿಮೆ ಮಾಡಿ, N95 ಮಾಸ್ಕ್ ಬಳಸಿ, ಮತ್ತು ಭಾರೀ ಕೆಲಸವನ್ನು ಪೀಕ್ ಸಮಯದ ನಂತರ ಮಾಡಿ.

### MARATHI

Delhi, Ward 12: पुढील 24 तासांत हवा खूप खराब राहण्याचा अंदाज आहे. बाहेरील हालचाल कमी ठेवा, N95 मास्क वापरा, आणि जड काम पीक वेळेनंतर करा.

### TAMIL

Delhi, Ward 12: அடுத்த 24 மணி நேரத்தில் காற்றின் தரம் மிக மோசம் இருக்கும் என எதிர்பார்க்கப்படுகிறது. வெளியில் செல்வதைக் குறைக்கவும், N95 முகக்கவசம் அணியவும், கடின வேலைகளை உச்ச நேரத்திற்குப் பிறகு செய்யவும்.

### TELUGU

Delhi, Ward 12: రాబోయే 24 గంటల్లో గాలి నాణ్యత చాలా చెడ్డదిగా ఉండవచ్చు. బయటి కార్యకలాపాలు తగ్గించండి, N95 మాస్క్ ధరించండి, భారీ పనిని పీక్ సమయం తర్వాత చేయండి.

### BENGALI

Delhi, Ward 12: আগামী 24 ঘণ্টায় বাতাস খুব খারাপ থাকার পূর্বাভাস। বাইরে কম সময় থাকুন, N95 মাস্ক পরুন এবং ভারী কাজ পিক সময়ের পরে করুন।

### GUJARATI

Delhi, Ward 12: આગામી 24 કલાકમાં હવા ખૂબ ખરાબ રહેવાની આગાહી છે. બહારની પ્રવૃત્તિ ઓછી રાખો, N95 માસ્ક પહેરો અને ભારે કામ પીક સમય પછી કરો.

Deterministic templates (no LLM), script-validated; app · Telegram bot · IVR line (Hindi voice for Hindi-first cities) · public displays; 5,495 vulnerability-scored zones (hospitals, schools, elder-care, outdoor work × population).

## All ten cities, both baselines — negatives kept

| city · window          | test hours @24 h | vs persistence 24/48/72 h | vs weekly seasonal-naive | 80 % PI coverage |
|------------------------|------------------|---------------------------|--------------------------|------------------|
| Delhi · multi-season   | 2,34,491         | +9% / +13% / +12%         | +26% / +20% / +16%       | 78% / 78% / 78%  |
| Mumbai · multi-season  | 1,41,667         | +17% / +19% / +21%        | +37% / +35% / +32%       | 82% / 82% / 79%  |
| Kolkata · multi-season | 59,373           | +14% / +10% / +9%         | +26% / +14% / +13%       | 75% / 73% / 70%  |
| Ahmedabad · live 90 d  | 689              | +12% / +13% / +14%        | +13% / +16% / +14%       | 71% / 78% / 80%  |
| Bengaluru · live 90 d  | 1,209            | +12% / +34% / +42%        | +11% / -13% / +1%        | 84% / 83% / 80%  |
| Chennai · live 90 d    | 732              | +5% / +5% / +6%           | +22% / +4% / +2%         | 82% / 82% / 80%  |
| Delhi · live 90 d      | 2,272            | +12% / +20% / +17%        | +30% / +27% / +22%       | 79% / 77% / 73%  |
| Hyderabad · live 90 d  | 595              | -5% / -1% / +21%          | +36% / +21% / +29%       | 75% / 72% / 72%  |
| Jaipur · live 90 d     | 428              | -16% / -10% / -31%        | +23% / -2% / -9%         | 75% / 74% / 47%  |
| Kolkata · live 90 d    | 303              | +15% / +16% / +24%        | +9% / +11% / +21%        | 80% / 77% / 87%  |
| Lucknow · live 90 d    | 615              | +26% / +18% / +32%        | +38% / +34% / +25%       | 75% / 69% / 78%  |
| Mumbai · live 90 d     | 0                | - / - / -                 | - / - / -                | - / - / -        |
| Pune · live 90 d       | 1,180            | +17% / +15% / +25%        | +24% / +26% / +29%       | 77% / 80% / 75%  |

Protocol: strict temporal split, rolling monthly refit on the trailing 90 days (= production), one shared support mask; served forecast = LightGBM median blended with persistence, conformal 80 % band, calibrated exceedance probabilities. TFT-lite trained on GPU and rejected. `python -m ml.eval.benchmark · GET /metrics/benchmark`

## Honest by construction

### The model abstains

No ML blame without out-of-sample skill ( $R^2 \geq 0.15$ ); otherwise cited chemical-signature priors, and the UI says which.

### Health text is LLM-free

Deterministic templates in eight scripts — a language model cannot hallucinate medical advice.

### Notices are drafts

Stamped pending officer authorisation; never auto-sent. Dispatch starts before/after measurement against the city's drift.

### Nothing fabricated

Impact figures return null over invented constants; satellite evidence is real imagery of the real site; study numbers are read from the primary PDFs.

### No socio-economic inputs

Priority = contribution  $\times$  exposure  $\times$  actionability  $\times$  confidence. No income or land-value feature anywhere in the pipeline.

### Auditable end to end

Per-node agent traces + an immutable status log (who, when, from  $\rightarrow$  to, finding) that outlives the record it describes.

## ₹0 today — a config file per city, a measured path forward

### Stack · ₹0

FastAPI on Render · React + MapLibre on Vercel  
· Supabase Postgres (PostGIS + pgvector) ·  
LangGraph · GitHub Actions crons. Free tier for  
the ten cities running today.

### Where ₹0 stops

| cities         | DB @180 d | ₹ / month     |
|----------------|-----------|---------------|
| 10 (today)     | ~465 MB   | 0             |
| 15             | ~655 MB   | ~2,100        |
| 50             | ~2.0 GB   | ~2,100        |
| 131 (all NCAP) | ~5.1 GB   | <b>~2,700</b> |

measured 0.21 MB/city/day; one row per reading; archive-first  
retention

### Security & roadmap

RLS on every table; public reads via anon JWT;  
writes only through the service role on the  
server; admin onboarding behind a key header.  
Roadmap: InMAP/PAVITRA source-receptor  
physics for the what-if engine, a learned  
satellite CV detector, more in-language IVR  
voices, permit-registry connectors.

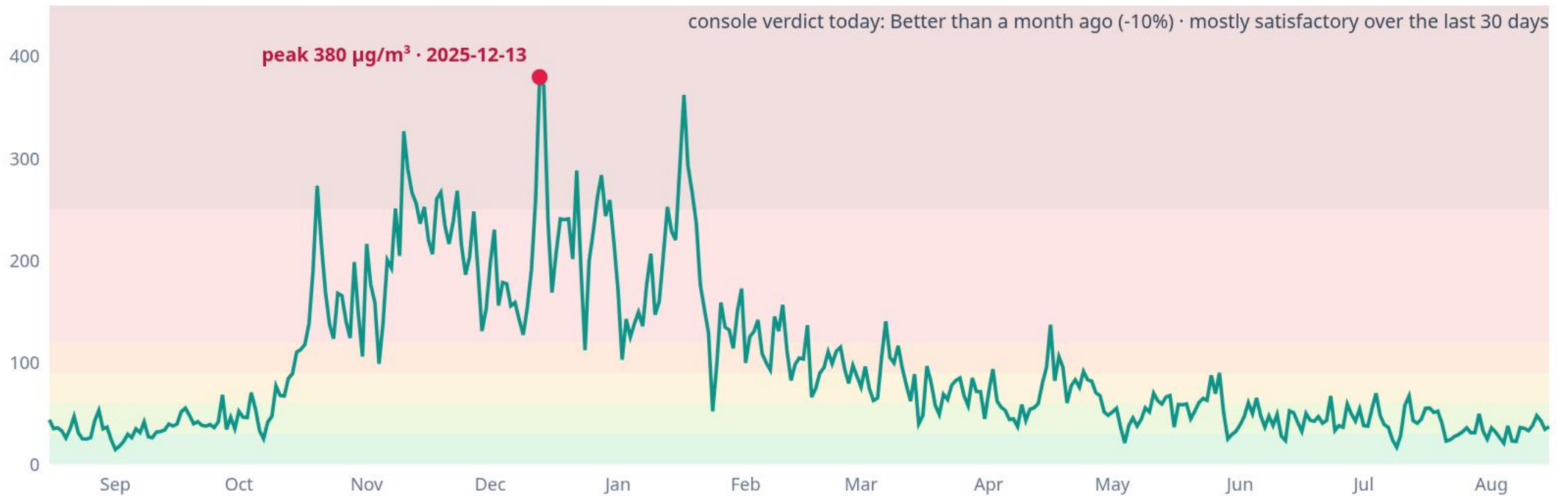
# One engine, ten cities — live



| #  | city      | PM2.5 now | +24 h | dominant source   | languages |
|----|-----------|-----------|-------|-------------------|-----------|
| 1  | Delhi     | 38.3      | 32.2  | construction dust | hi en     |
| 2  | Jaipur    | 34.4      | 34.2  | construction dust | hi en     |
| 3  | Ahmedabad | 32.6      | 25.6  | construction dust | gu en hi  |
| 4  | Lucknow   | 26.3      | 19.0  | industrial        | hi en     |
| 5  | Chennai   | 25.3      | 21.5  | construction dust | ta en     |
| 6  | Bengaluru | 22.0      | 11.4  | construction dust | kn en     |
| 7  | Pune      | 19.8      | 16.0  | construction dust | mr en     |
| 8  | Kolkata   | 19.1      | 22.5  | traffic           | bn en hi  |
| 9  | Mumbai    | 13.2      | 11.8  | traffic           | mr en hi  |
| 10 | Hyderabad | 11.0      | 18.6  | traffic           | te en hi  |

live comparison · as of 19 Aug, 10:39 pm

## Delhi, one year of daily PM2.5 — the real winter



station-day means over 39 Delhi stations (CPCB via OpenAQ) — the same record the forecast benchmark uses; CPCB bands · the console shows the same view from its live table U archived rollout (GET /history/trend)

# India can measure and forecast its air. The gap is acting on it — audited.

## What exists

SAFAR (IITM, 4 cities, city-level bulletin) · AQEWS / DSS (WRF-Chem, 400 m grid, 29-source — CAQM/government users only) · PRANA (cities report NCAP actions & spend) · NCAP Tracker · CAAQMS / SAMEER (a reading, not a decision).

## What the audit found — CEEW, Oct 2025

'Severe and above' caught 5 of 14 episodes (1 of 15 the winter before) · PM2.5 under-predicted by  $\sim 24 \mu\text{g}/\text{m}^3$  in winter · CAQM imposed GRAP III/IV on *observed* AQI, not on forecasts · DSS lacks outcome monitoring and actionable short-term pathways.

## What VayuNetra adds

Calibrated P(>band) on every cell · onset recall reported vs persistence · live hourly attribution, not a 2016 inventory · forecast → named notice with citations → dispatch → measured outcome, closed with a finding · public, 8 languages, IVR/Telegram/web · ₹0 on the sunk CAAQMS investment.

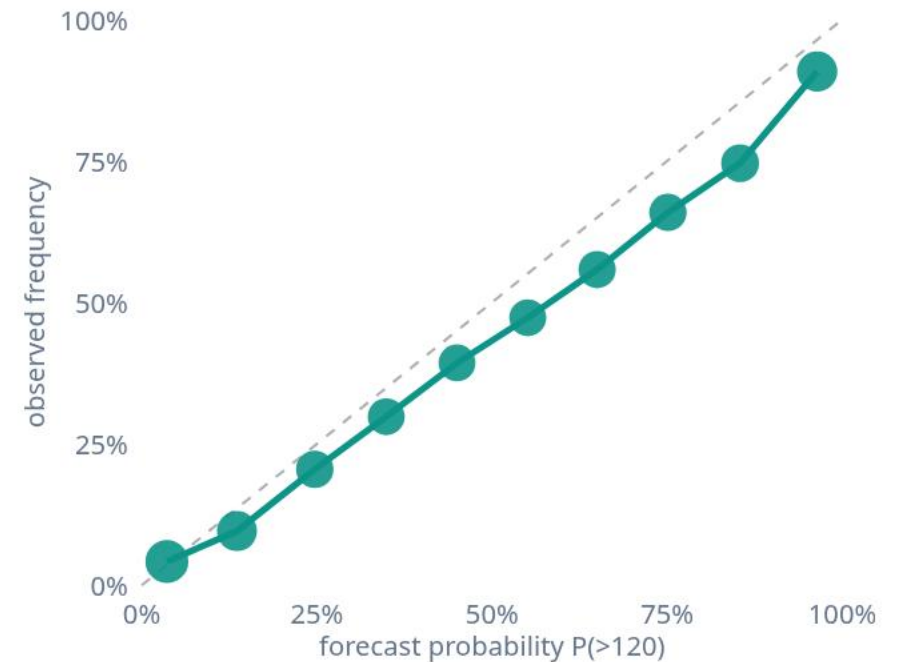
We do not claim to out-model WRF-Chem. Sources: CEEW 'How Well can Delhi Predict Air Quality?' (Ignatious & Rafiuddin, 2025); Ghude et al., GMD 17 (2024); docs/POSITIONING.md

## Does it warn before the air turns? Measured on 2025-26 Delhi

| horizon | onsets (>120) | onset recall — alarm on $P \geq 0.3$ | onset recall — persistence | alarm F1 ( $P \geq 0.3$ ) / persist. | $P(>120)$ Brier skill | 80 % PI coverage |
|---------|---------------|--------------------------------------|----------------------------|--------------------------------------|-----------------------|------------------|
| +24 h   | 15,701        | 54%                                  | 0%                         | 0.765 / 0.747                        | +51%                  | 78%              |
| +48 h   | 18,226        | 54%                                  | 0%                         | 0.744 / 0.708                        | +46%                  | 78%              |
| +72 h   | 19,117        | 51%                                  | 0%                         | 0.725 / 0.696                        | +38%                  | 78%              |

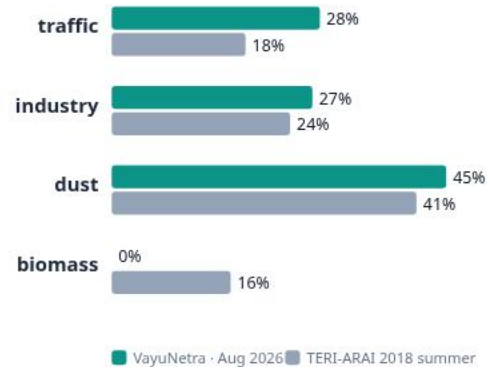
**Onset** = clean at issue time, above  $120 \mu\text{g}/\text{m}^3$  (Very Poor) at  $t+h$ . 'Tomorrow = today' has onset recall 0 by construction. **Where we are weak** — the Severe tail ( $>250$ ): the blended forecast only matches persistence there and the median alarm sees few Severe onsets (as the official WRF-Chem does); the product speaks in calibrated probability:  $P(>250)$  Brier skill +31 % at 24 h.

Reliability,  $P(>120)$ , +24 h

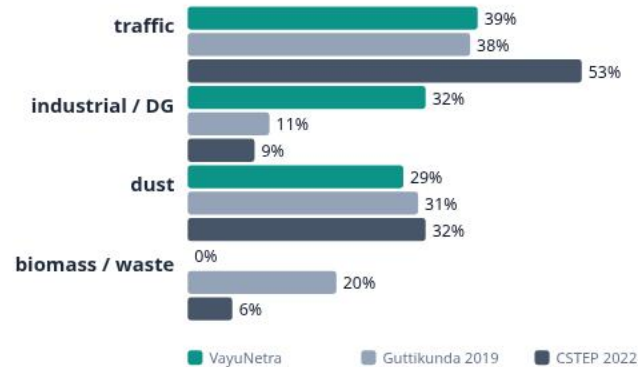


# Checked against published apportionment — agreements and disagreements both shown

## Delhi · source-only shares



## Bengaluru · source-only shares



## Who is in the forecast?

+24 h · Very Poor      **0.1% · ~11,837 people**

+48 h · Very Poor      **0.1% · ~9,980 people**

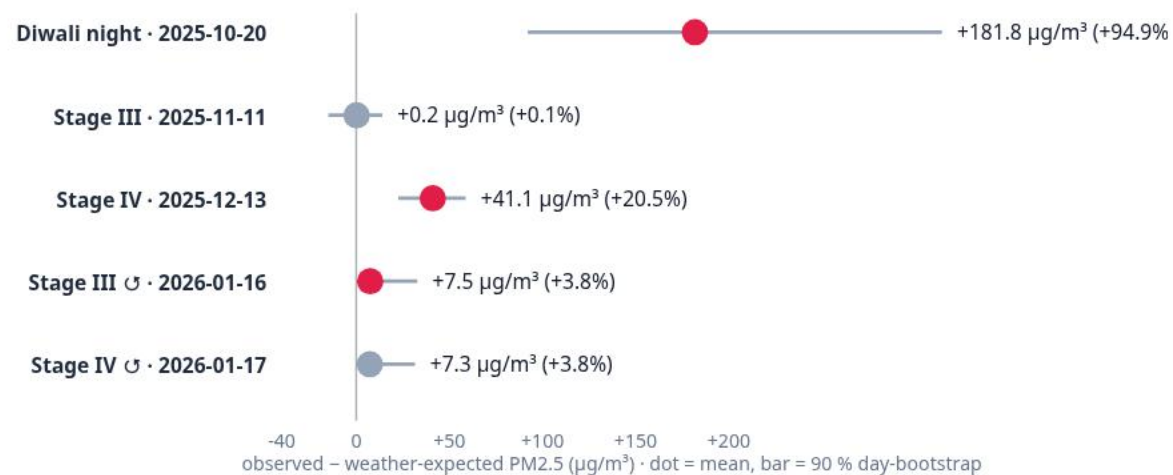
+72 h · Very Poor      **0.1% · ~19,580 people**

basis: gpw411\_cells (1,50,516 people in sampled cells; city-scaled figure assumes the same share city-wide — the card says so)

$\Sigma$  cell population  $\times$  calibrated  $P(> \text{band})$ ; GPW v4.11 where sampled, cited city population otherwise. Exposure, not mortality. GET /exposure

Read honestly: rankings agree; we over-read kerbside traffic (arterial-road stations,  $\text{NO}_2/\text{CO}$  signatures) and book less as regional than a dispersion model; our Bengaluru 'industrial' is DG sets, our biomass misses small waste fires. Shares move between daily runs where  $\text{NO}_2$  coverage is thin — shown, not smoothed. docs/ATTRIBUTION\_VALIDATION.md

## Did the air change during the order, weather taken out?



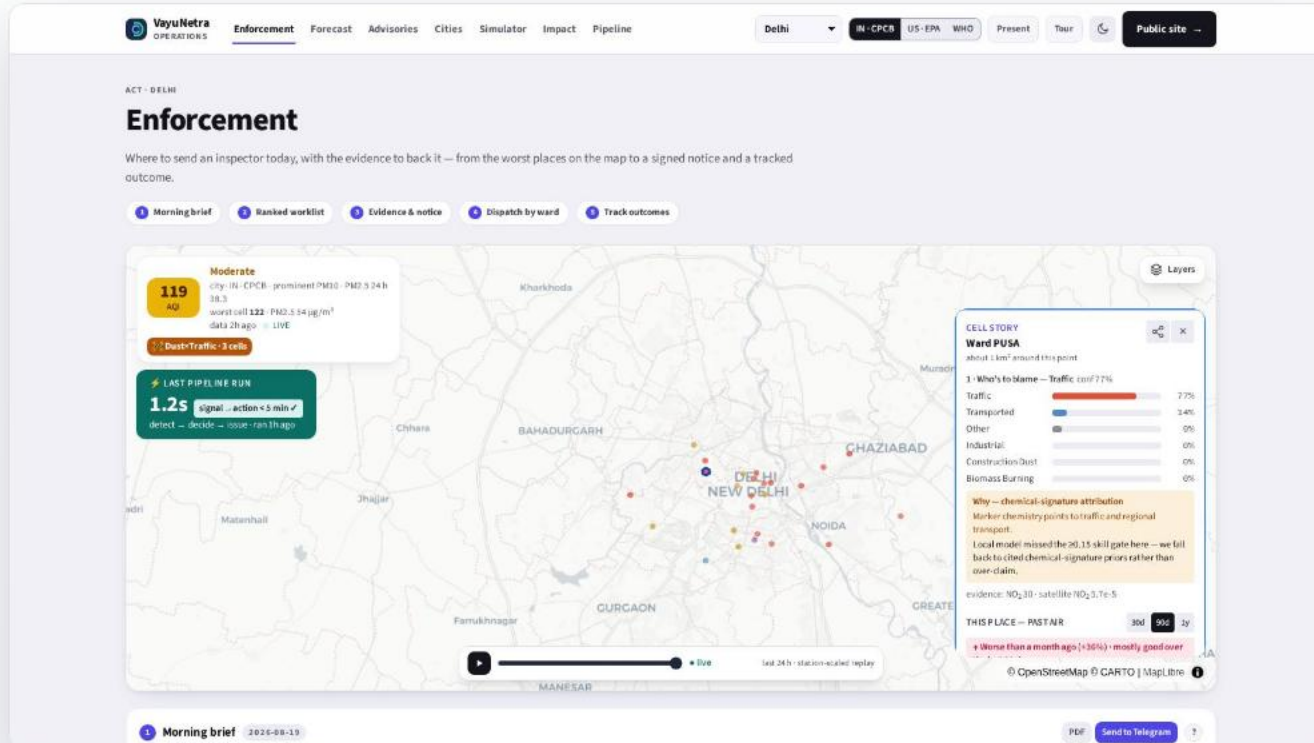
LightGBM on ERA5 meteorology + hour/dow/doy/cell, fitted on 88,270 season hours outside the windows; held-out  $R^2$  0.61, RMSE 65 µg/m³. python -m ml.eval.interventions

### Read honestly

Diwali night shows the method has the power to see a large signal. For the GRAP windows we find **no weather-adjusted reduction this method can detect**; the Stage IV rows sit in the most stagnant weather of the season, where a tree model fitted on calmer hours under-predicts. Association, not causation — coincident factors stay in the number.

This is why VayuNetra measures each dispatched action against **its own cell and the city's drift** — a targeted action has a control; a blanket stage does not.

# See → decide → act → prove — with an audit trail



1 **Approve** — moves the item to its ward queue

2 **Dispatch** — freezes the cell's 7-day baseline; effect vs city drift is measured from then on

3 **Close with a finding** — violation found · compliant · inaccessible · n/a, plus a note

4 **History** — immutable log: from → to, who, when, finding; survives the nightly run and even deletion of the record

**PRANA-ready CSV** — every tracked action with baseline / after / effect / status / closure, mapped to its NCAP head.